

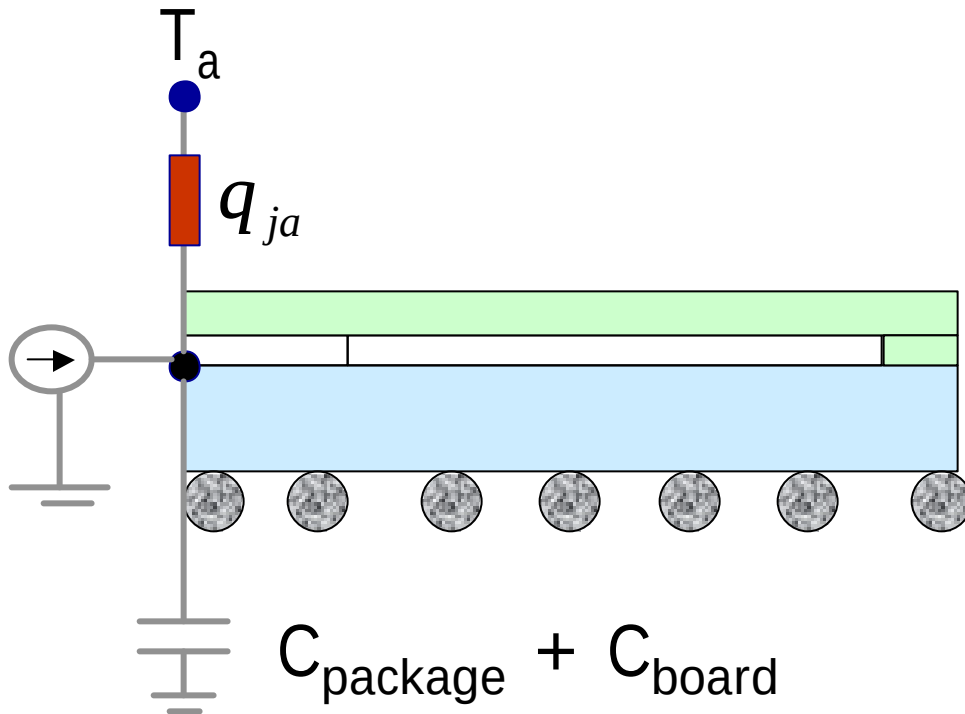


BACKGROUND

- ◆ What are the goals?
 - Model a package as a network of pseudo-resistors and pseudo-capacitors
 - Accurately simulate transient behavior independent of surface boundary conditions
 - Expand the Steady State methodology into the transient domain
- ◆ Questions
 - What topologies are suitable for transient simulation?
 - How and where are the capacitors introduced?
 - What is the effect of time intervals in parameter extraction?
 - Is it possible to achieve BC independence?



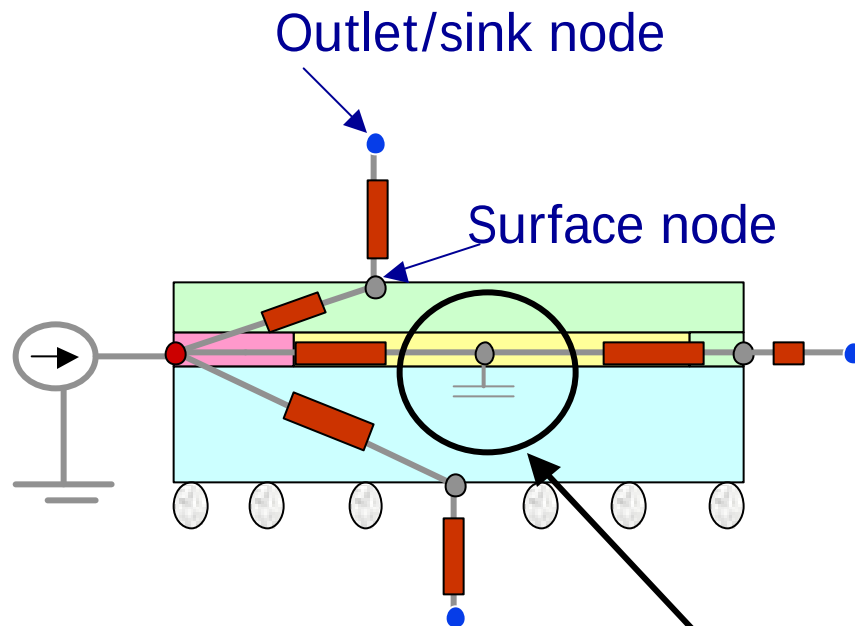
SIMPLEST RC MODEL REPRESENTATION



- ◆ Model Deficiencies
 - Steady State q_{ja} models are poor predictive models
 - Highly boundary condition dependent
 - There are too few degrees of freedom to capture the early *and* late time behavior
 - Internal and external resistance effects are coupled in q_{ja}



STAR-SHAPED MODEL REPRESENTATION

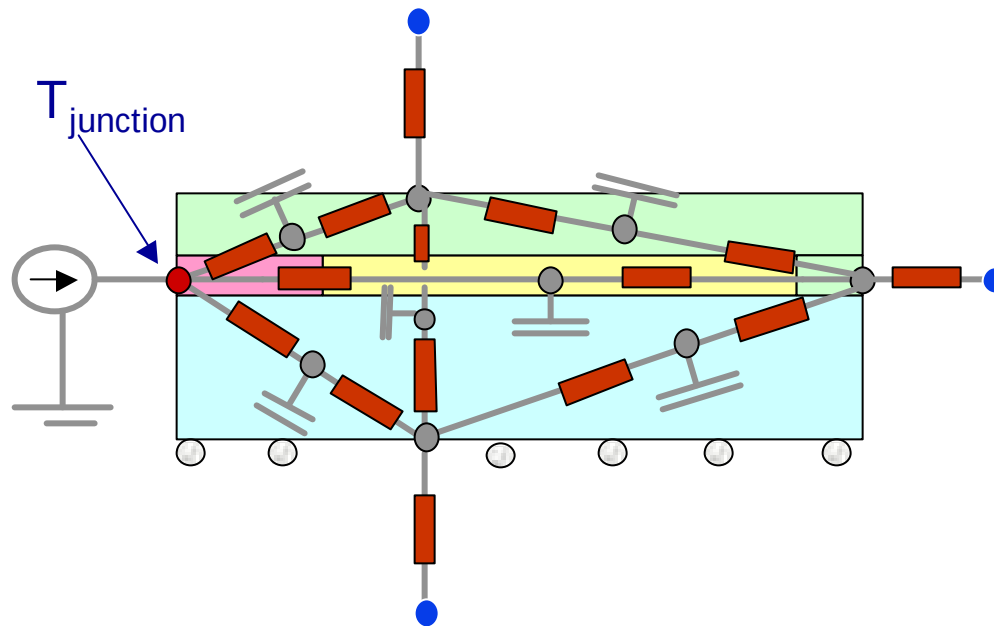


- ◆ Improvement over the previous model due to the increased DOF
- ◆ Steady State Star-shaped models lack accuracy
- ◆ Placement of capacitance is unclear

Capacitor is arbitrarily placed half way between 2 resistors



FULLY-CONNECTED NETWORK



- ◆ Added shunt resistance (surface to surface)
- ◆ Additional links are known to increase performance in Steady State
- ◆ Addition and placement of capacitance is under investigation



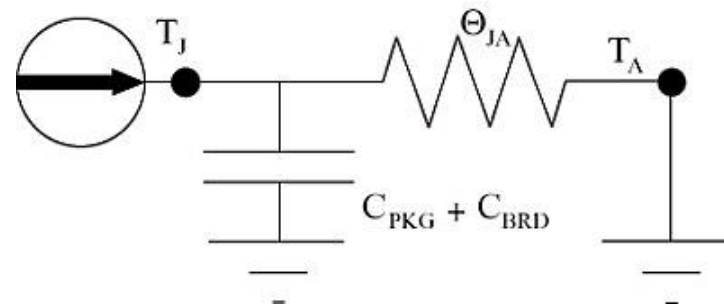
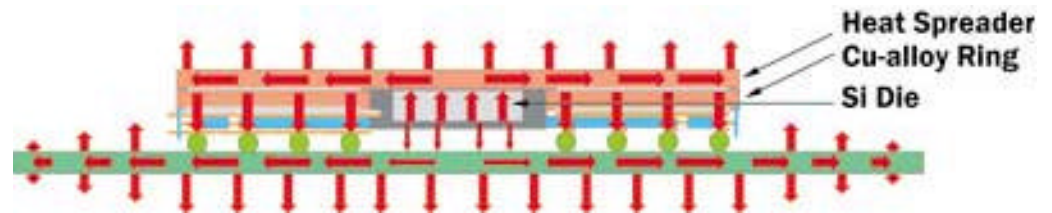
REVIEW OF PREVIOUS WORK

- ◆ Guenin (2002)
 - Traditional physics based RC model
 - Serially connected
 - Prespecify resistances and lumped masses
- ◆ Christiaens et al. (1998)
 - Parameter extraction and optimization approach (DELPHI)
 - Resistances are steady state values
 - Capacitances are optimized to minimize error in junction temperatures between compact and fully detailed models
- ◆ Rencz et al. (2001)
 - Electrical circuit multi-port theory



1 STAGE Q_{JA} MODEL GUENIN (2002)

- ◆ 3D BGA package
- ◆ Obtained temperature response curve experimentally
- ◆ Represents 3D package with 1D serially connected circuit





SOLUTION: 1 STAGE Q_{JA} GUENIN (2002)

- ◆ The standard solution for 1 stage RC network:

$$T_j(t) = \Delta T_{j,steadystate} * (1 - e^{-t/\tau})$$

$$\Delta T_{j,steadystate} = Power * \Theta_{ja}$$

$$\tau = R * C = \Theta_{ja} * C_{total}$$

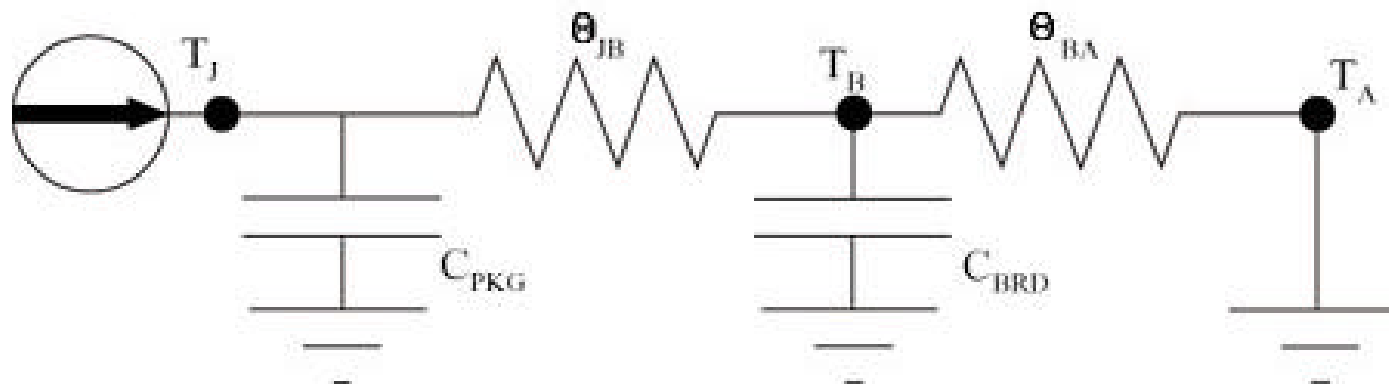
$$C_{total} = C_{package} + C_{board}$$

- ◆ R and C values were chosen a priori
- ◆ Engineer must be very knowledgeable to use this approach as 1-D semi-analytic solutions are used



2 STAGE RC NETWORK GUENIN (2002)

- ◆ R and C values chosen a priori using lumped mass techniques





SOLUTION: 2 STAGE RC NETWORK

GUENIN (2002)

$$T_j(t) = \Delta T_{j,ss} \cdot \left[\frac{\Psi_{jB}}{\Theta_{jA}} \cdot (1 - e^{-t/t_1}) + \frac{\Psi_{BA}}{\Theta_{jA}} \cdot (1 - e^{-t/t_2}) \right]$$

$$\Psi_{jB} = \frac{\Delta T_{jB}}{Power}$$

$$\Psi_{BA} = \frac{T_{board} - T_{air}}{Power}$$

$$\Theta_{jA} = \Psi_{jB} + \Psi_{BA}$$

Time Constant Information

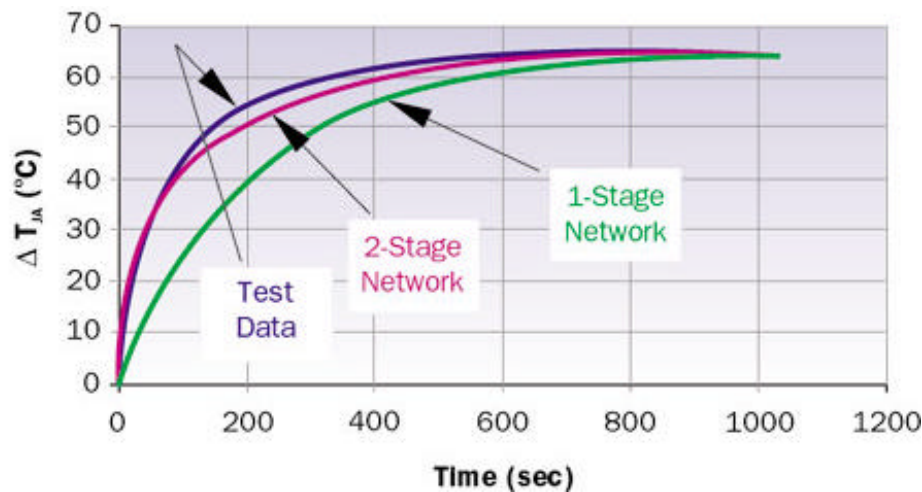
$$t_1 = \Psi_{jB} \cdot C_{package}$$

$$t_2 = \Theta_{jA} (C_{package} + C_{board})$$

$$t_2 \gg t_1$$



COMPARISON OF 1 STAGE & 2 STAGE SOLUTIONS TO DATA GUENIN (2002)

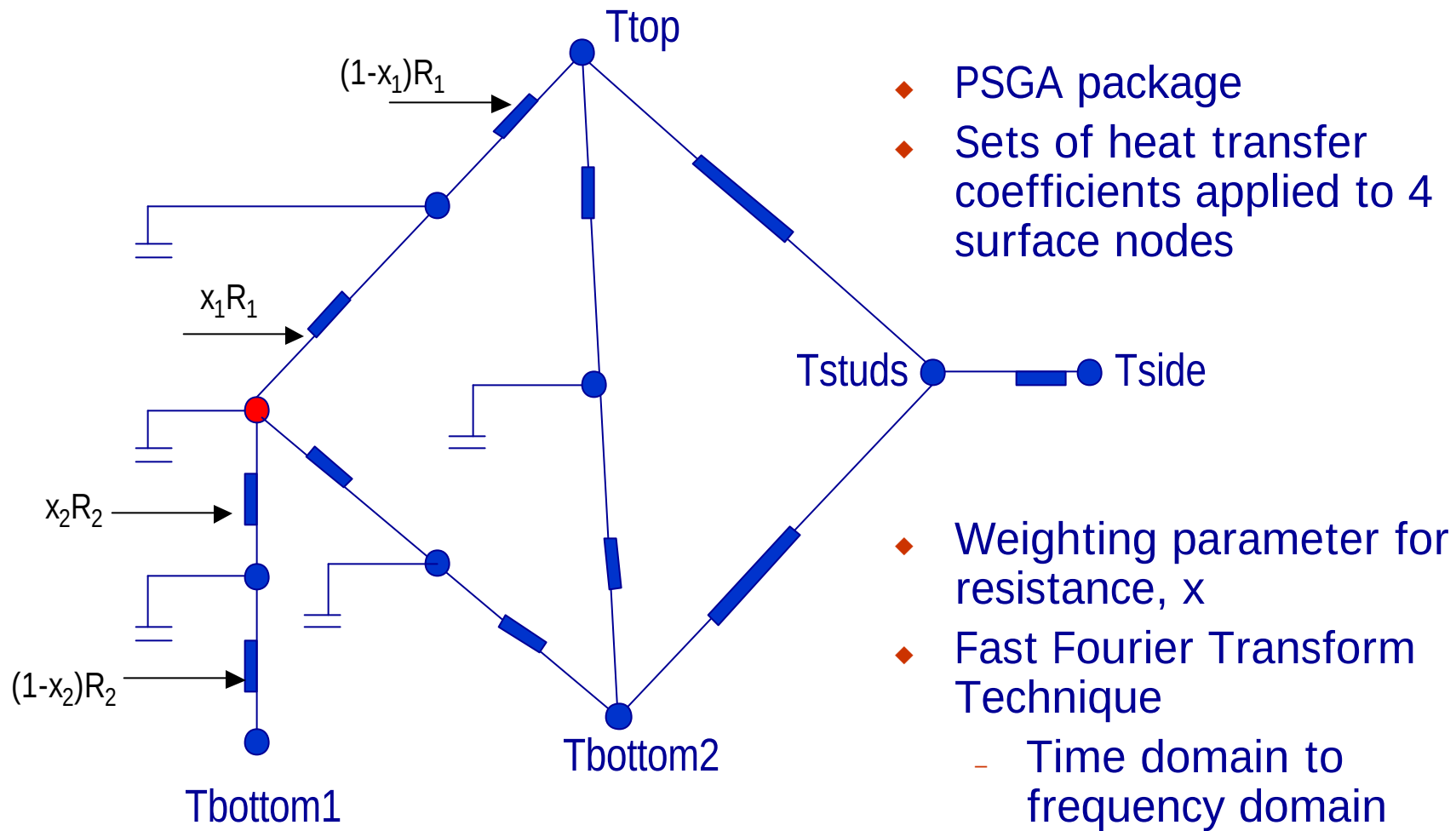


- ◆ Clearly 2-stage model performs better than 1-stage model
- ◆ 2-stage model shows improvement for both early and late time behavior
- ◆ T_{ja} is not accurately predicted if the time interval is less than a few seconds
- ◆ R and C values are specified a priori by using physical models
- ◆ 3D package can be accurately represented as a 1D problem



TRANSIENT EXTENSION OF THE DELPHI MODEL

CHRISTIAENS ET AL. (1998)





TRANSIENT EXTENSION OF THE DELPHI MODEL

CHRISTIAENS ET AL. (1998)

- ♦ Minimize cost function:

$$\sum_{i=1}^N \sum_{k=1}^n \left(\frac{|q_j^{FEM,i}(w_k)| - |q_j^{CM,i}(w_k)|}{|q_j^{FEM,i}(w_k)|} \right)^2$$

n = number of angular frequency abscissa

N = number of BC for which accurate temperature response data is available

$|q_j^{FEM,i}(w_k)| =$ magnitude of the Fourier transform of the simulated junction temperature step response for BC i and frequency w_k

$|q_j^{CM,i}(w_k)| =$ magnitude of the Fourier transform of the junction temperature step response predicted by the compact model for BC i and frequency w_k



OBSERVATIONS

- ◆ Applies heat transfer coefficients to 4 sides of package
 - 6 different sets of BCs
- ◆ Resistors are steady state values
- ◆ Optimization performed by MATLAB
- ◆ Solution is represented as 7 resistors, 4 capacitors, 4 x coefficients
- ◆ Solution is boundary condition independent



MOTIVATION FOR PRESENT WORK

- ◆ Explore statistical parameter extraction methodology into the transient domain
- ◆ Develop methodology for BC independent transient models
- ◆ Explore the effect of time interval divisions on model parameters
- ◆ Explore the effect of various topologies on accuracy
- ◆ Develop UACTM-Transient

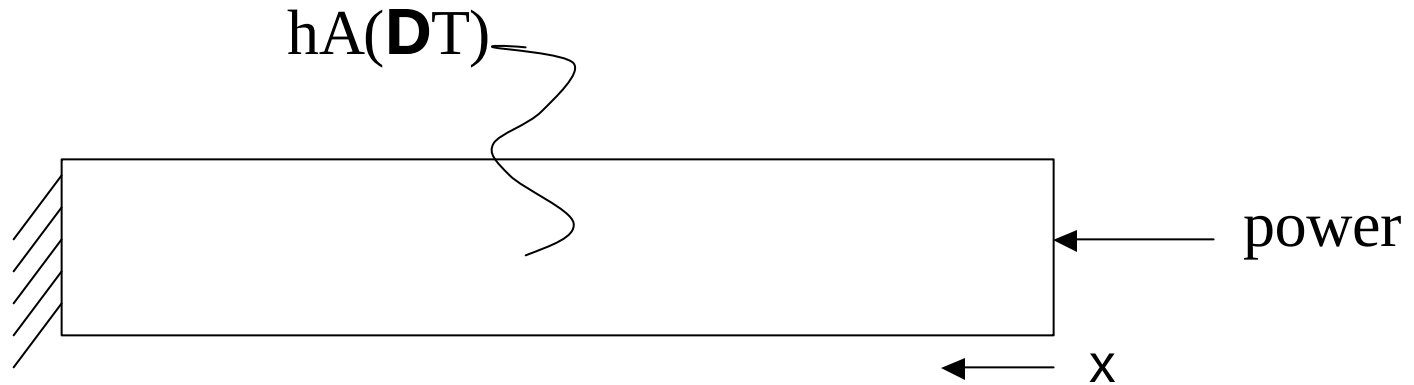


APPROACH

- ◆ Survey existing methodologies
- ◆ Formulate optimization methodology for a 1D transient problem with known solution (transient fin problem)
- ◆ Apply methodology to a BGA package (Guenin's simple 1D topology)
- ◆ Extend transient CTM for fully-connected topology



TRANSIENT FIN PROBLEM



◆ Formulation:

$$q_t = q_{xx} - b * q$$

$$q(x, 0) = 0$$

$$q_x(0, t) = 0$$

$$q_x(1, t) = 1$$



ANALYTICAL SOLUTION

◆ Solution:

$$q(x, t) = \frac{x^2}{2} + \left(\frac{1}{b} - \frac{1}{6} \right) - \frac{e^{-bt}}{b} + \sum_{n=1}^{10} A(t, n) f(x, n)$$

$$A(t, n) = \frac{2(-1)^n}{l_n} \left[\left(\frac{b}{l_n + b} - 1 \right) e^{-(l_n + b)t} - \frac{b}{l_n + b} \right]$$

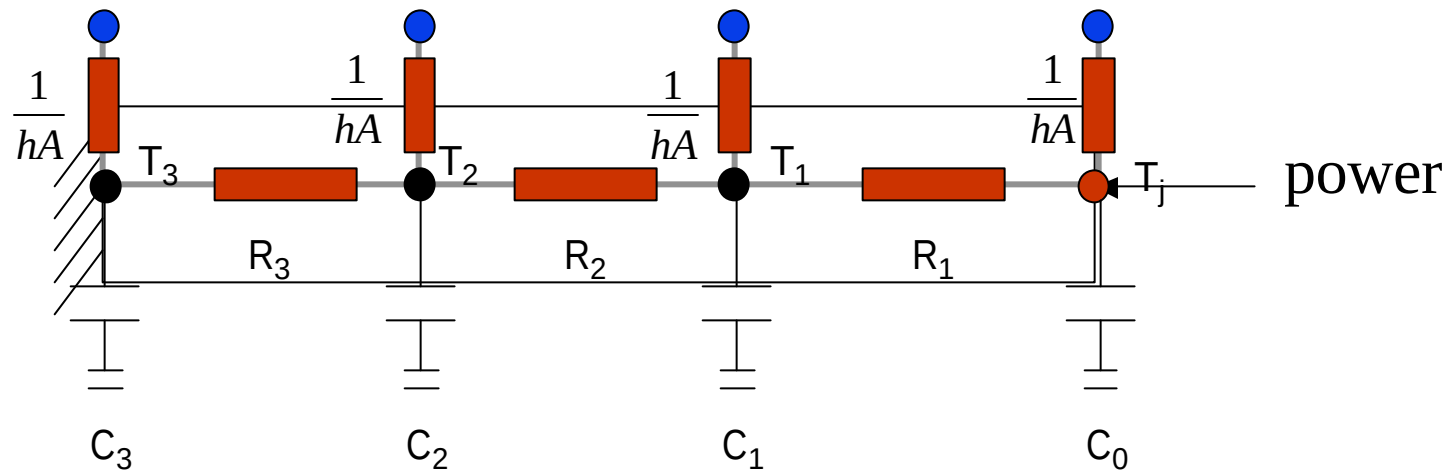
$$f_n(x) = \cos(\sqrt{l_n} x)$$

$$l_n = (np)^2$$

$$b = \frac{hP}{kA}$$



NODAL ENERGY EQUATIONS



$$T_1^{n+1} = T_1^n + \frac{\Delta t}{C_1} \left(\frac{T_j^n - T_1^n}{R_1} + \frac{T_2^n - T_1^n}{R_2} + hA(T_1 - T_a) \right)$$



NODAL ENERGY EQUATIONS

- ◆ Write nodal equations for each node

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_j^{n+1} \\ T_1^{n+1} \\ T_2^{n+1} \\ T_3^{n+1} \end{bmatrix} = \begin{bmatrix} T_j^n + \frac{\Delta t}{C_0} \left(q + \frac{T_1^n - T_j^n}{R_1} + h_0 A_0 (T_j^n - T_a) \right) \\ T_1^n + \frac{\Delta t}{C_1} \left(\frac{T_j^n - T_1^n}{R_1} + \frac{T_2^n - T_1^n}{R_2} + h_1 A_1 (T_1^n - T_a) \right) \\ T_2^n + \frac{\Delta t}{C_2} \left(\frac{T_1^n - T_2^n}{R_2} + \frac{T_3^n - T_2^n}{R_3} + h_2 A_2 (T_2^n - T_a) \right) \\ T_3^n + \frac{\Delta t}{C_3} \left(\frac{T_2^n - T_3^n}{R_3} + h_3 A_3 (T_3^n - T_a) \right) \end{bmatrix}$$

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COST FUNCTION

Minimize:

$$CostFunction = \sum \left(\frac{T_j^c - T_j^f}{T_j^f - T_{ambient}} \right)^2$$

Such that:

$$Error = \left(\frac{T_j^c - T_j^f}{T_j^f - T_{ambient}} \right) * 100$$

$$-10\% < Error < 10\%$$

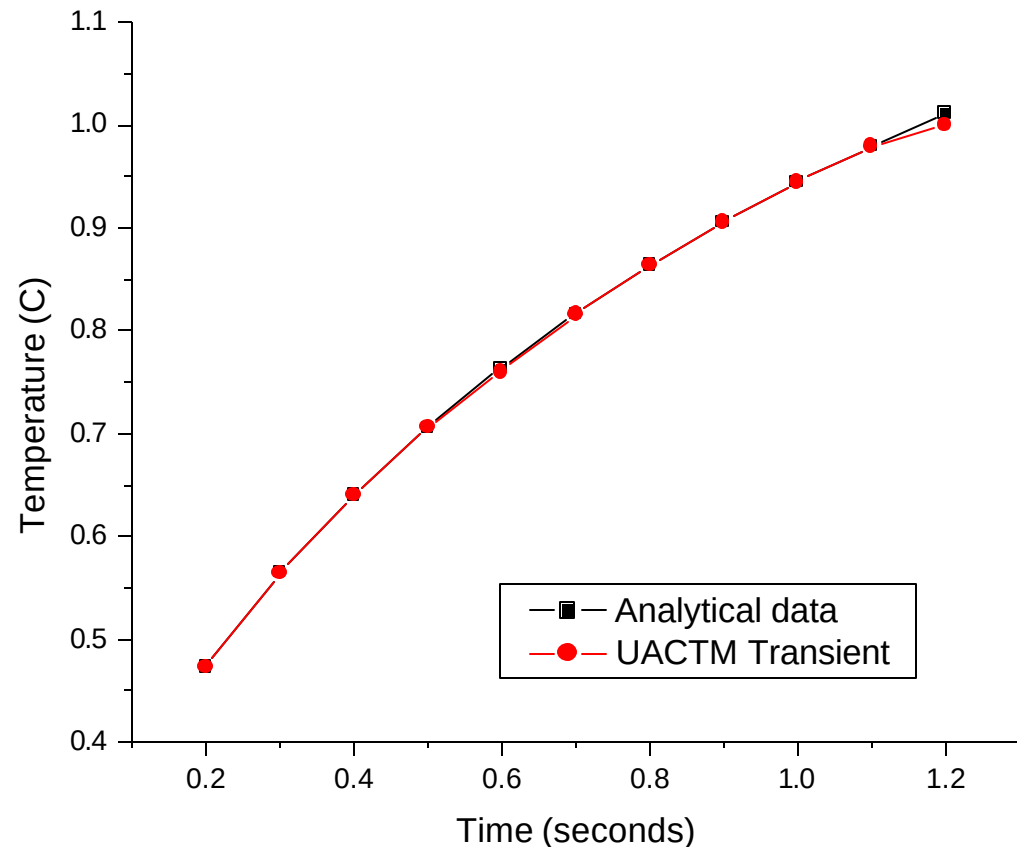
$$0.0001 < \text{Resistance values} < 1000000$$

$$0.0001 < \text{Capacitance values} < 1000000$$



OBSERVATIONS

- ◆ Represented the rod as 3 resistors and 4 capacitors
- ◆ Values of R and C are not dependent on time interval
- ◆ The final solution depends on starting values for R and C
- ◆ Looked at ways to express resistance and capacitance as functions of length
- ◆ R and C do not satisfy any physical constraint
- ◆ Validation step for methodology



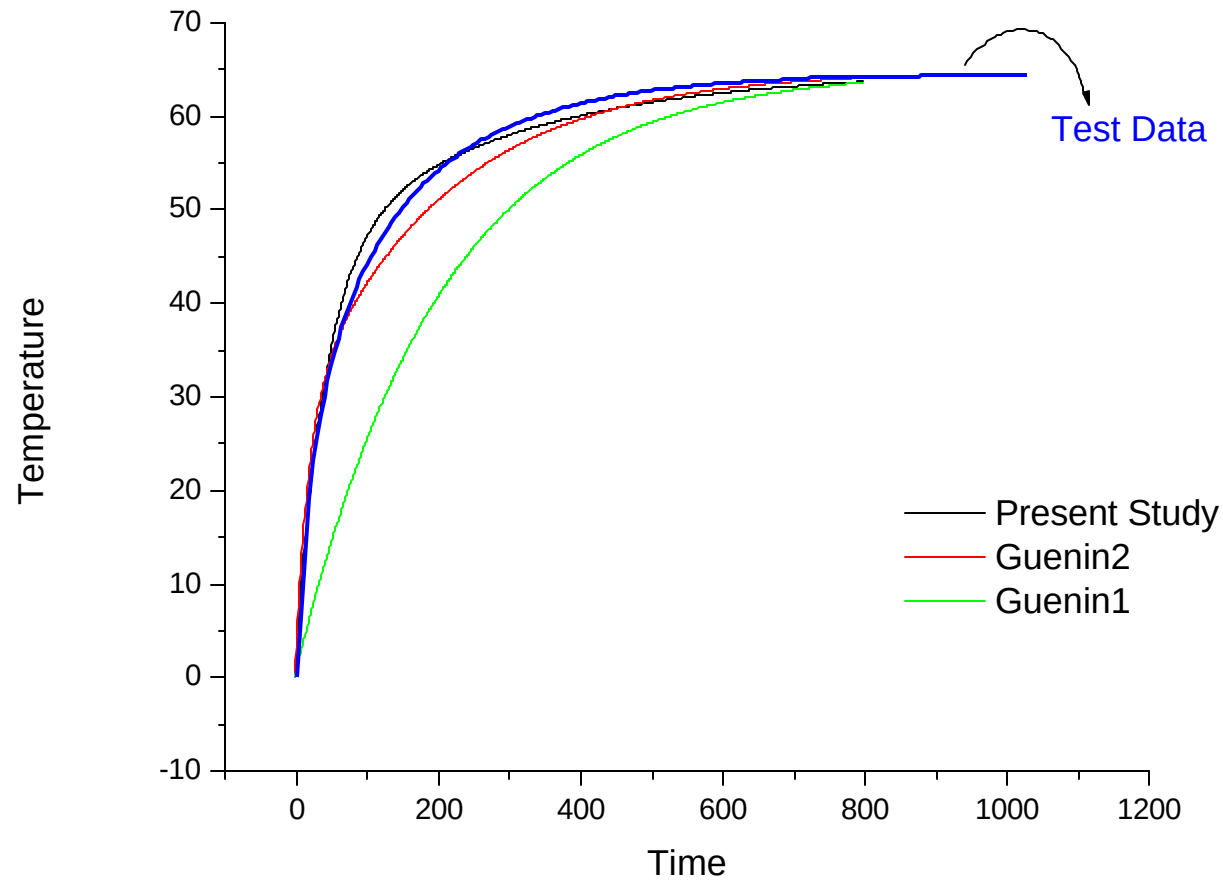


TRANSIENT CTM FOR 1D BGA PACKAGE

- ◆ Nodal Energy Equations – 3 nodes, 2 resistors, 2 capacitors
- ◆ Guenin's model used as reference case
- ◆ Two approaches
 - Variable R, constant C
 - Variable R, variable C
- ◆ R and C not strongly dependent on time interval used for extraction



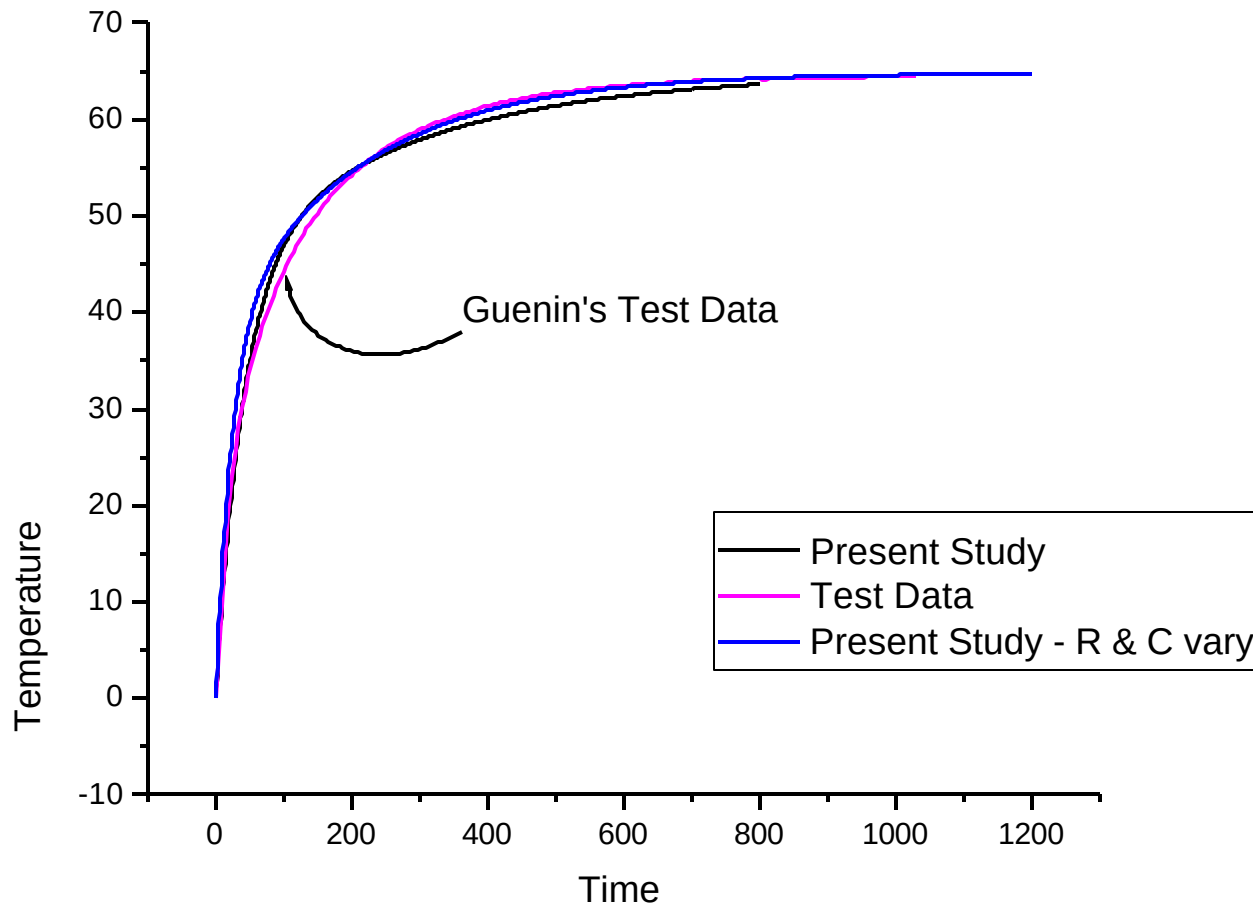
RESULTS



- ◆ Guenin1:
1-Stage solution
- ◆ Guenin2:
2-Stage solution
- ◆ Present Study:
Variable R,
constant C



RESULTS



- ◆ Gabel:
Variable R,
constant C
- ◆ Gabel:
Variable R,
Variable C
- ◆ Methodology
provides an
accurate
result



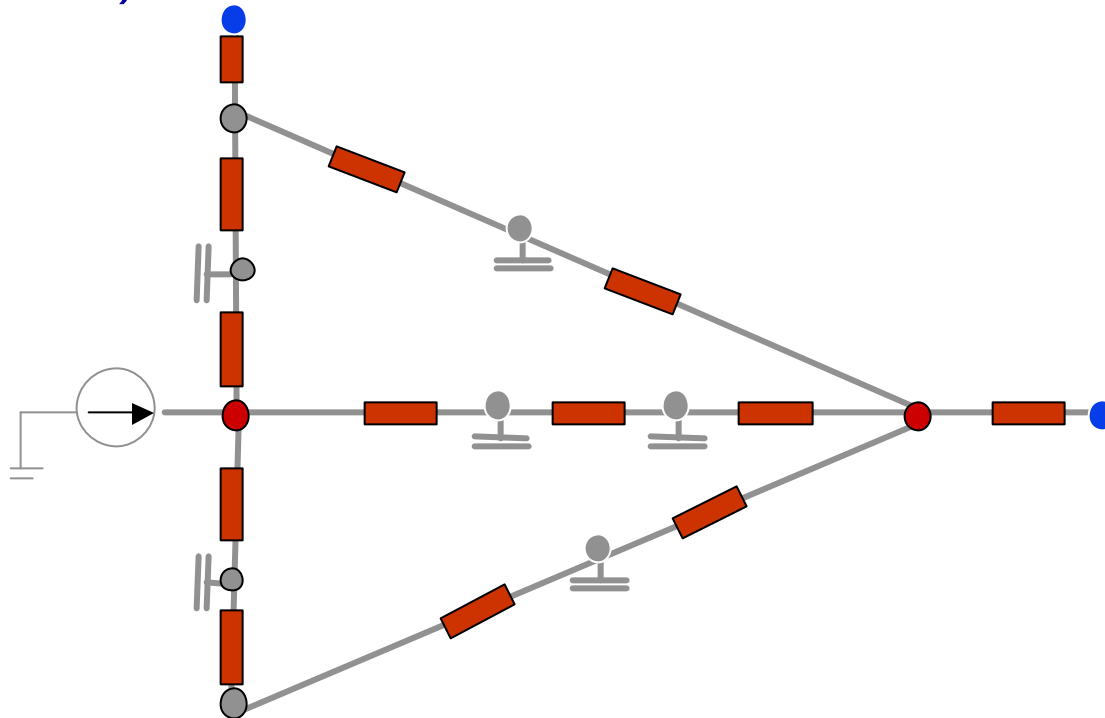
CONCLUSION

- ◆ Summary of other methodologies was briefly presented
- ◆ Proposed methodology applied to 2 cases
 - 1D fin problem
 - 1D representation of 3D BGA package
- ◆ Application of nodal energy equations in the transient domain
- ◆ Optimization of parameters
- ◆ Solution does not depend on the specified time interval
 - Solution is non-unique



CONTINUING WORK

- ◆ 3-D Models in ANSYS
 - CPGA and BGA
- ◆ Investigation of optimum topologies for transient models
- ◆ Customized optimization methods (J. C. Smith, University of Arizona)





Recent References

1. Christiaens F., Vandeveld B., Beyne E., Mertens R., and Jberghmans J. [“A Generic Methodology for Deriving Compact Dynamic Thermal Models, Applied to the PSGA Package,”](#) *IEEE Trans. Comp., Packag., Manufact. Technol. A*, vol. 21, no. 4, pp. 565-576, Dec. 1998.
2. Guenin, Bruce. [“Simplified Transient Model for IC Packages,”](#) Aug. 2002.
http://www.electronics-cooling.com/html/2002_august_calccorner.html
3. Rencz, M., Szekely, V. [“A Generic Method for Thermal Mltiport Model Generation of IC Packages,”](#) Seventeenth IEEE SEMI-THERM Symposium, pp. 145-152, March 2001.